

A VISUAL LEARNING HANDBOOK

Networking

From packets and secure connections to modern APIs, real-time streams, and identity.

20 ESSENTIAL TOPICS

MINIMAL FLOW
CHARTS

PRODUCTION INTUITION

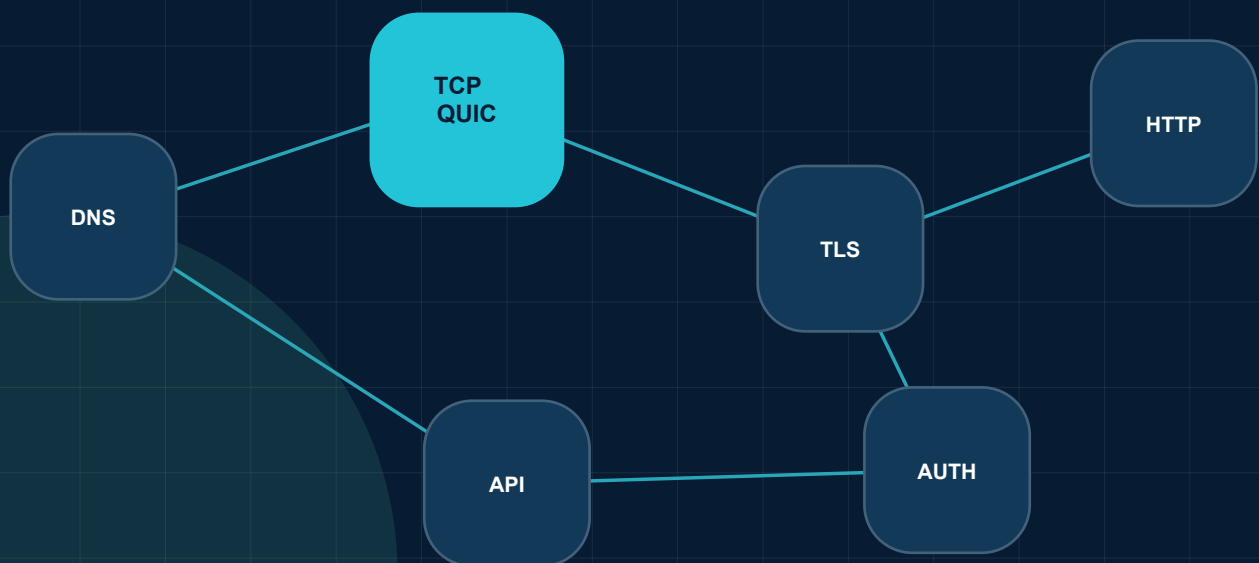


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Follow the stack from transport and web protocols through edge delivery, APIs, real-time communication, browser security, and identity.

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HOW TO READ

Each topic is one page. Read the summary, trace the flow, then review the core ideas and decision boxes.

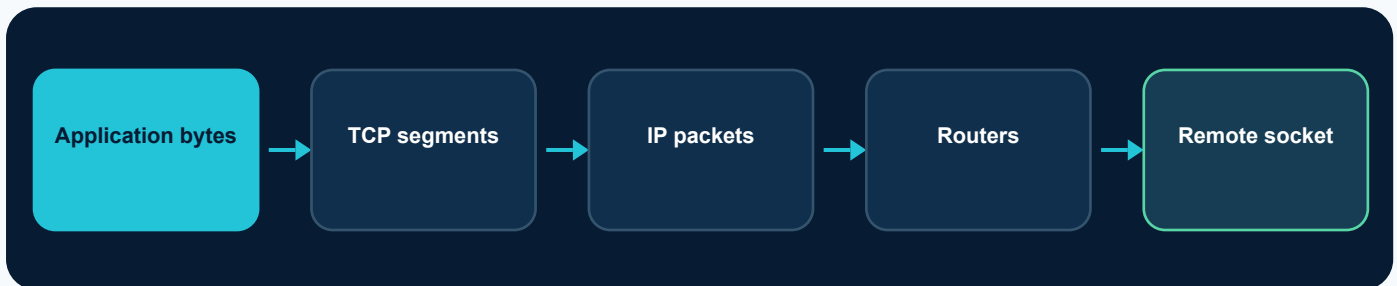
TCP/IP Protocol Suite

Reliable byte streams across an unreliable packet network.

IN ONE MINUTE

IP routes packets between hosts. TCP builds a connection above IP and turns those packets into an ordered, reliable, congestion-aware byte stream.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Connection setup

SYN, SYN-ACK, and ACK synchronize sequence numbers and confirm two-way reachability.

2

Reliable delivery

Sequence numbers, ACKs, checksums, and retransmission repair loss and reordering.

3

Traffic control

Receive and congestion windows protect the receiver and adapt to network capacity.

MAKE THE DECISION

USE IT WHEN

Web traffic, databases, SSH, email, and workloads where complete ordered delivery matters.

WATCH OUT FOR

Head-of-line delay, retransmission latency, connection storms, and exhausted socket tables.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

TCP guarantees an ordered byte stream, not message boundaries and not zero latency.

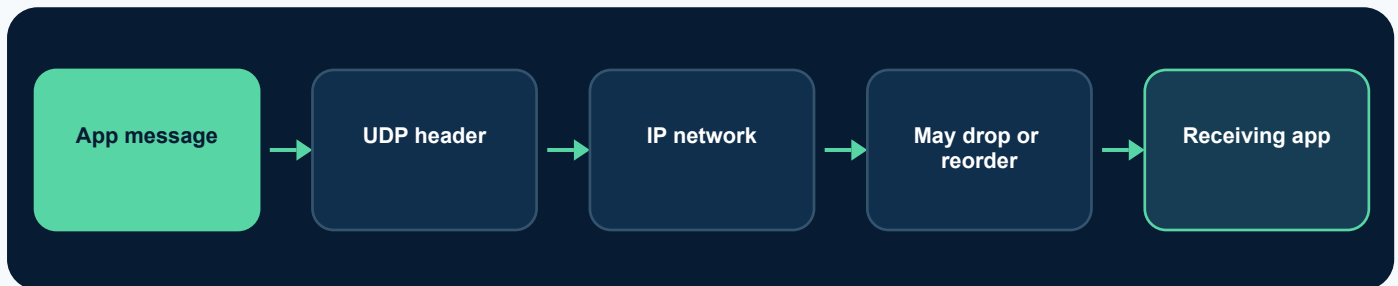
UDP

Small independent datagrams with almost no transport ceremony.

IN ONE MINUTE

UDP adds ports and a checksum to IP, then sends each message without a handshake. Delivery, ordering, retries, and pacing belong to the application.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Connectionless

A sender can transmit immediately without creating shared connection state.

2

Message boundaries

Each datagram remains a discrete message rather than joining a byte stream.

3

Best effort

Loss, duplication, and reordering can occur, so the application chooses its recovery policy.

MAKE THE DECISION

USE IT WHEN

DNS, voice/video, gaming, telemetry, discovery, and latency-sensitive updates.

WATCH OUT FOR

Amplification, fragmentation, unbounded send rates, silent loss, and custom reliability bugs.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

UDP starts quickly because it promises little; reliability is a design choice above it.

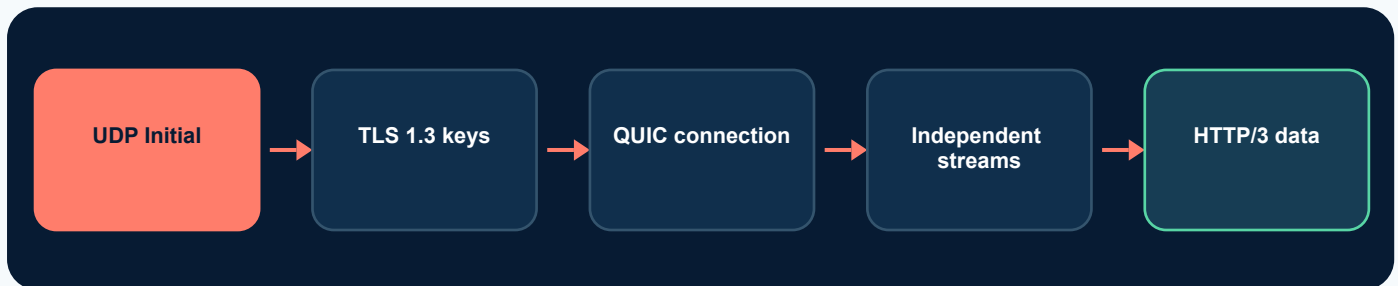
QUIC

Encrypted multiplexed transport built in user space over UDP.

IN ONE MINUTE

QUIC combines TLS 1.3, reliable streams, congestion control, and connection migration. Its streams recover independently instead of sharing one TCP byte order.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Integrated security

Transport and TLS negotiation happen together; nearly all packet content is encrypted.

2

Stream isolation

Loss pauses the affected stream while unrelated complete data can still be delivered.

3

Connection IDs

A logical identifier allows a connection to survive an IP or port change.

MAKE THE DECISION

USE IT WHEN

HTTP/3, mobile clients, lossy networks, and multiplexed latency-sensitive traffic.

WATCH OUT FOR

UDP blocking, 0-RTT replay, CPU cost, observability changes, and difficult middleboxes.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

QUIC keeps reliability inside each stream instead of forcing all streams through one byte order.

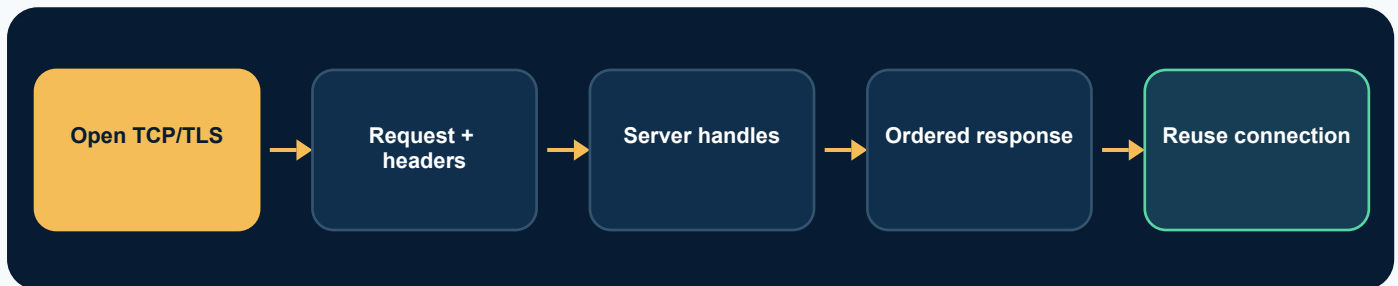
HTTP/1.1

Readable request and response messages over persistent TCP.

IN ONE MINUTE

HTTP/1.1 defines methods, status codes, headers, and bodies. Keep-alive reuses a connection, but mainstream clients cannot safely multiplex many responses on it.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Text messages

Request and response lines are readable and followed by an optional body.

2

Persistent connection

Keep-alive amortizes TCP and TLS handshakes across multiple exchanges.

3

Limited parallelism

Browsers use connection pools because one connection lacks robust multiplexing.

MAKE THE DECISION

USE IT WHEN

Simple services, broad compatibility, proxies, and modest request concurrency.

WATCH OUT FOR

Head-of-line delay, ambiguous framing, connection limits, and unsafe retries.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

Keep-alive saves handshakes, but it does not give HTTP/1.1 true multiplexing.

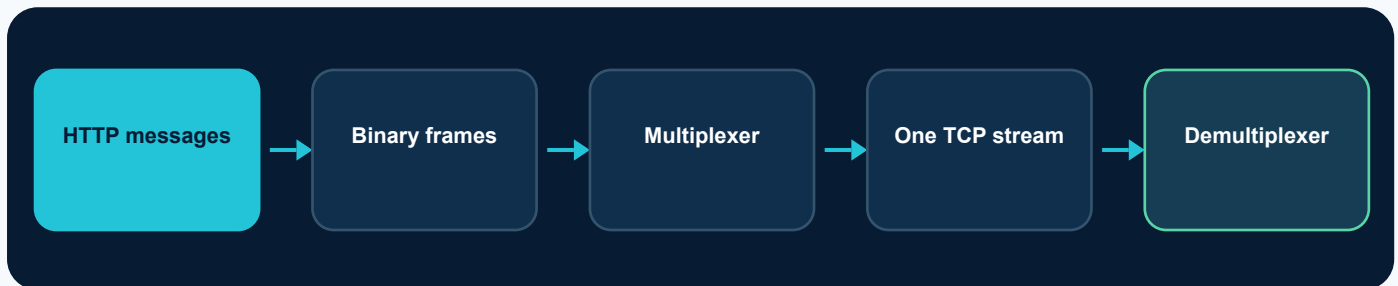
HTTP/2

Binary frames multiplex many HTTP streams over one TCP connection.

IN ONE MINUTE

HTTP/2 preserves HTTP meaning while encoding messages as frames. Stream IDs interleave requests and responses, and HPACK compresses repeated headers.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Binary framing

HEADERS and DATA frames carry message parts with explicit lengths and flags.

2

Multiplexing

Many logical streams share one connection without HTTP/1.1 response ordering.

3

TCP dependency

A lost TCP segment delays later bytes for every stream on the connection.

MAKE THE DECISION

USE IT WHEN

Browsers, API gateways, gRPC, and origins serving many concurrent resources.

WATCH OUT FOR

TCP head-of-line blocking, flow-control mistakes, and single-connection overload.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

HTTP/2 removes HTTP-level blocking but cannot remove TCP-level ordered delivery.

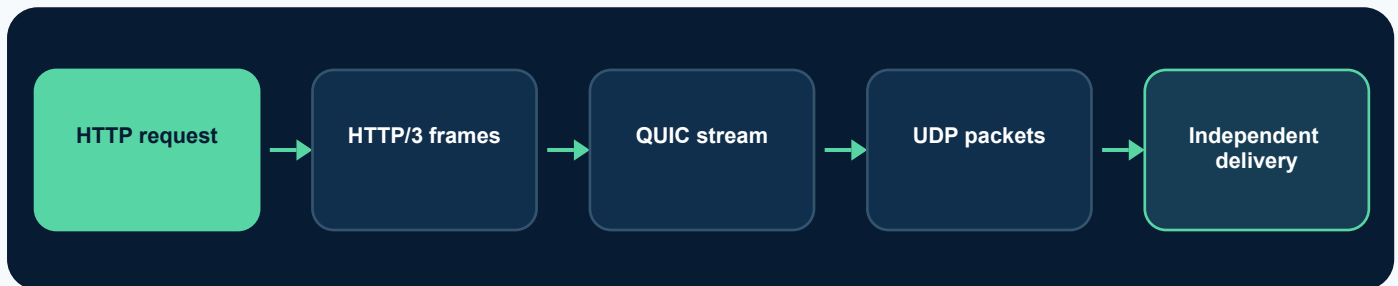
HTTP/3

HTTP semantics mapped onto QUIC streams instead of TCP.

IN ONE MINUTE

HTTP/3 uses QUIC for secure transport. Request streams recover independently, while QPACK compresses headers without one globally ordered byte stream.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

QUIC transport

Encryption, stream control, recovery, and migration are integrated below HTTP.

2

No cross-stream HoL

Loss on one stream does not block complete data on unrelated streams.

3

QPACK

Dedicated control streams coordinate header compression while limiting blocking.

MAKE THE DECISION

USE IT WHEN

Modern public web delivery, mobile networks, and sites sensitive to loss-induced latency.

WATCH OUT FOR

Fallback behavior, UDP filtering, edge support, CPU cost, and new monitoring needs.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

HTTP/3 changes transport behavior, not familiar HTTP methods, status codes, or URLs.

HTTPS

HTTP protected by authenticated encryption and server identity.

IN ONE MINUTE

HTTPS carries HTTP inside TLS. Certificates authenticate the hostname, key exchange derives shared secrets, and symmetric keys protect requests and responses.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Confidentiality

Observers can see some metadata but cannot read protected HTTP contents.

2

Integrity

Authenticated encryption detects modification or forged TLS records.

3

Authentication

The browser validates the certificate chain and requested hostname.

MAKE THE DECISION

USE IT WHEN

Every public website, API, login, payment flow, and untrusted network path.

WATCH OUT FOR

Expired certificates, mixed content, weak redirects, leaked keys, and missing validation.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

HTTPS protects data in transit; it does not make application logic trustworthy.

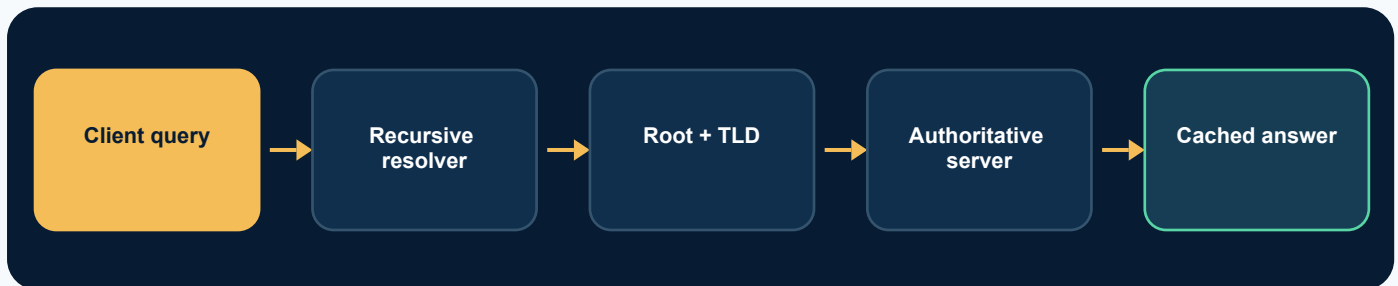
DNS

A distributed hierarchy that maps names to service information.

IN ONE MINUTE

DNS lets clients find services by name. Recursive resolvers follow referrals to authoritative servers, then cache answers according to TTL.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Hierarchy

Root delegates to TLDs, which delegate domains to authoritative nameservers.

2

Record types

A/AAAA map addresses; CNAME, MX, NS, TXT, and PTR serve other roles.

3

Caching and TTL

Caches trade faster answers and resilience for slower change propagation.

MAKE THE DECISION

USE IT WHEN

Service discovery, global routing, mail delivery, and human-friendly addressing.

WATCH OUT FOR

Stale caches, dangling records, poisoning, DDoS, and slow failure propagation.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

DNS chooses where a connection starts; it does not move application data.

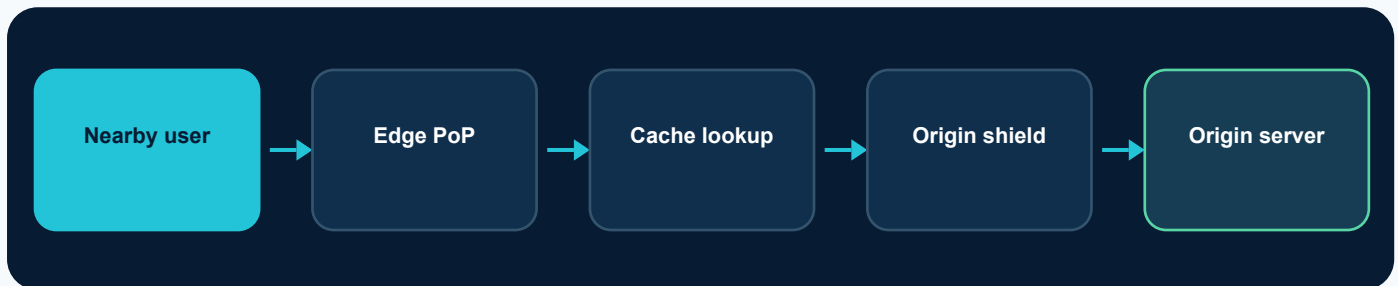
Content Delivery Network

Serve content close to users and protect the origin from repeated work.

IN ONE MINUTE

A CDN places cache and proxy nodes at global points of presence. Nearby edges serve fresh objects and forward misses through shields or the origin.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Edge caching

Popular objects travel fewer network miles and avoid repeated origin work.

2

Freshness policy

Cache-Control and validators decide reuse, revalidation, and expiration.

3

Origin shielding

A regional cache collapses simultaneous misses before they reach the origin.

MAKE THE DECISION

USE IT WHEN

Static assets, video, downloads, public APIs, global sites, and DDoS absorption.

WATCH OUT FOR

Wrong cache keys, private data leaks, stale content, stampedes, and purge delay.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

A CDN is a distributed reverse proxy; cache correctness matters as much as hit rate.

SSL/TLS

Authenticate peers, agree on keys, and protect application bytes.

IN ONE MINUTE

TLS creates a secure channel over a transport. TLS 1.3 uses ephemeral key exchange, certificate signatures, transcript checks, and authenticated encryption.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Certificate chain

A trusted CA vouches that a public key is authorized for the hostname.

2

ECDHE

Ephemeral Diffie-Hellman derives shared secrets without sending them on the wire.

3

Forward secrecy

A later long-term key compromise should not decrypt captured past sessions.

MAKE THE DECISION

USE IT WHEN

HTTPS, databases, service meshes, internal services, and zero-trust traffic.

WATCH OUT FOR

Old versions, weak ciphers, broken trust stores, failed rotation, and clock skew.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

TLS secures a channel only after identity and certificate validation succeed.

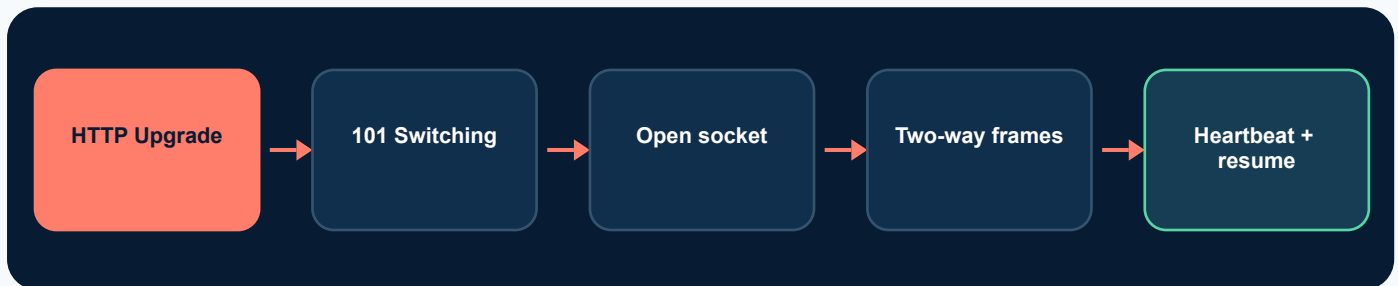
WebSocket

A long-lived full-duplex message channel initiated through HTTP.

IN ONE MINUTE

WebSocket begins with an HTTP Upgrade and then switches to compact frames. Client and server can push messages independently on one persistent connection.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Full duplex

Either peer can send at any time without opening another HTTP request.

2

Connection state

Servers track sockets, subscriptions, authentication, and backpressure.

3

Horizontal scale

A broker fans events across nodes because each socket belongs to one process.

MAKE THE DECISION

USE IT WHEN

Chat, collaborative editing, trading screens, multiplayer state, and control channels.

WATCH OUT FOR

Idle timeouts, reconnect storms, slow consumers, fan-out, and missing replay.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

A live socket is not durable messaging; add IDs and replay when loss is unacceptable.

Server-Sent Events

Simple server-to-browser streaming over a long HTTP response.

IN ONE MINUTE

SSE sends UTF-8 event records with text/event-stream. Browsers reconnect automatically and can report the last received event ID.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

One direction

The server pushes events; client commands use ordinary HTTP requests.

2

Text framing

Blank lines separate events and fields carry data, type, ID, or retry delay.

3

Auto reconnect

EventSource retries and sends Last-Event-ID when one is available.

MAKE THE DECISION

USE IT WHEN

Notifications, dashboards, logs, progress, feeds, and token-by-token AI output.

WATCH OUT FOR

Proxy buffering, HTTP/1.1 connection limits, replay storage, and slow clients.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

Choose SSE when updates flow mainly server to client and text events are enough.

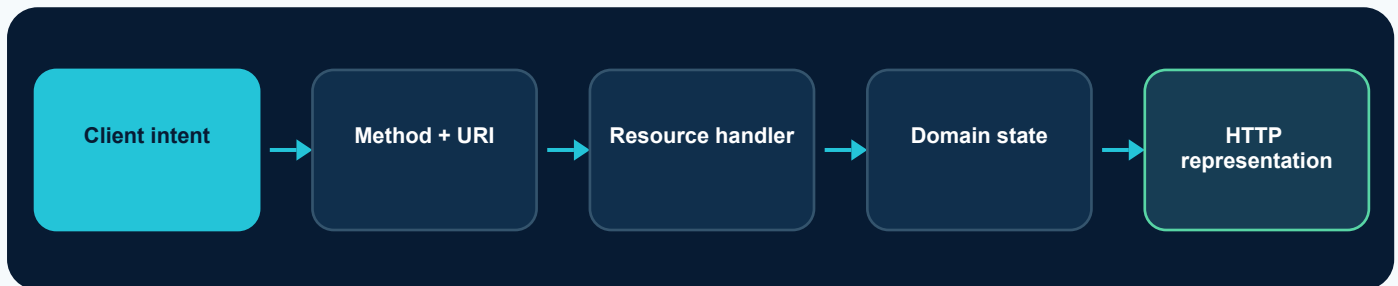
REST API

Resource-oriented interfaces built from standard HTTP semantics.

IN ONE MINUTE

REST models business concepts as resources identified by URIs. Methods, representations, status codes, caching, and stateless requests create a uniform interface.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Resources

Nouns and stable identifiers describe what the client is interacting with.

2

HTTP semantics

GET is safe; PUT and DELETE are idempotent; status codes communicate outcomes.

3

Caching

Cache-Control, ETag, and conditional requests reduce transfer and lost updates.

MAKE THE DECISION

USE IT WHEN

Public APIs, CRUD-like resources, browser integration, and cache-friendly services.

WATCH OUT FOR

Action-heavy URLs, inconsistent errors, unsafe retries, and chatty designs.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

REST quality comes from consistent semantics and contracts, not from JSON alone.

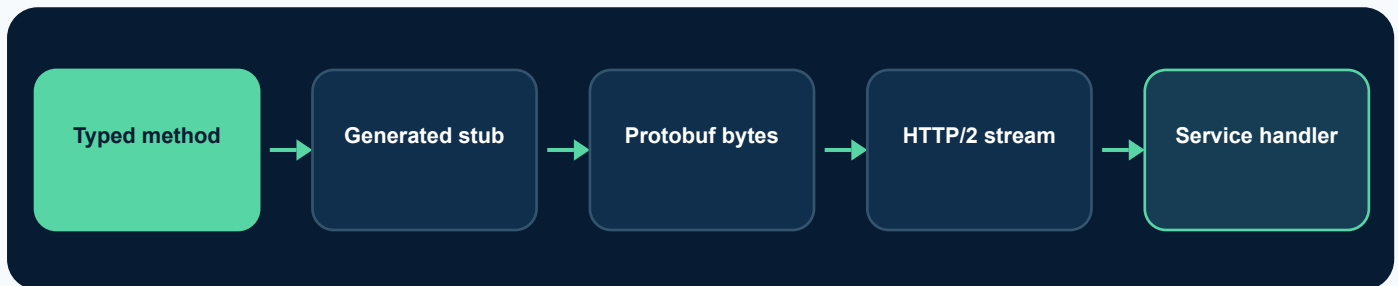
gRPC

Typed remote calls using Protobuf messages over HTTP/2.

IN ONE MINUTE

gRPC generates client and server stubs from a service contract. Compact Protobuf messages travel over HTTP/2 as unary or streaming calls.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Contract first

.proto files define services, methods, fields, and stable numeric tags.

2

Four call shapes

Unary, server-streaming, client-streaming, and bidirectional streaming share one model.

3

Deadlines

Clients state how long work remains useful; cancellation should propagate.

MAKE THE DECISION

USE IT WHEN

Internal microservices, polyglot systems, typed APIs, and streaming pipelines.

WATCH OUT FOR

Browser limits, schema compatibility, retry duplication, and missing deadlines.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

Treat a remote call as unreliable network work even when it looks like a local method.

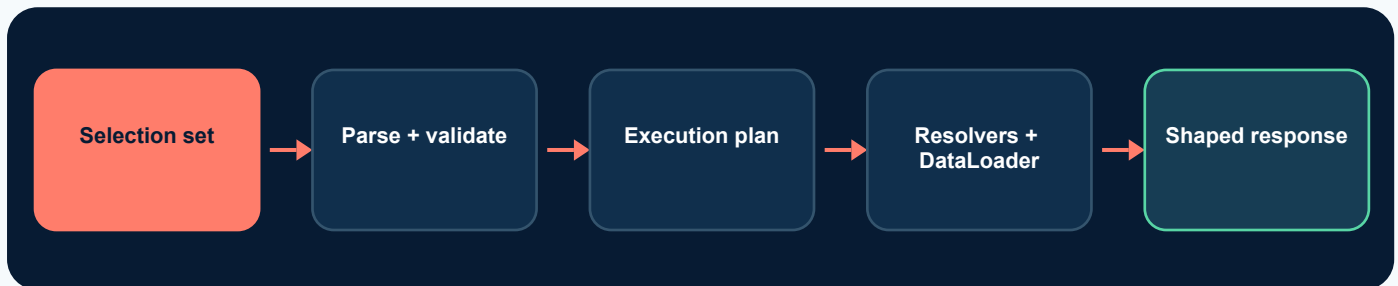
GraphQL

Clients declare the exact shape of data they need from a typed graph.

IN ONE MINUTE

GraphQL exposes a schema and usually one endpoint. The server validates each operation, runs field resolvers, and shapes a response to the selection set.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Typed schema

Objects, fields, arguments, and nullability form a discoverable contract.

2

Resolver execution

Each field can load from services or databases and may run in parallel.

3

Cost control

Depth, complexity, timeouts, persisted queries, and quotas contain abuse.

MAKE THE DECISION

USE IT WHEN

Product UIs with varied data needs, graph-like domains, and evolving client views.

WATCH OUT FOR

N+1 queries, field leaks, unbounded complexity, cache difficulty, and partial errors.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

GraphQL moves response flexibility to clients and execution complexity to servers.

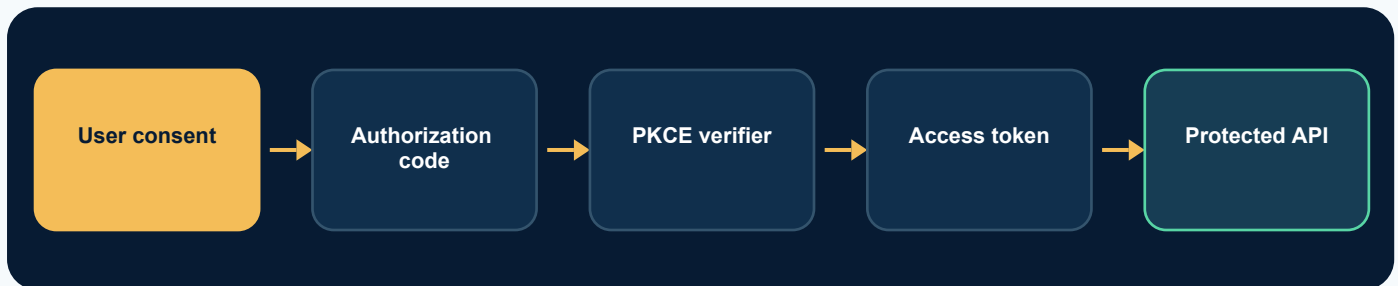
OAuth 2.0 and OIDC

Delegate API access without giving an application the user's password.

IN ONE MINUTE

OAuth 2.0 issues scoped access tokens to clients. OpenID Connect adds identity claims and an ID token so a client can establish who authenticated.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Separate roles

Owner, client, authorization server, and resource server cross trust boundaries.

2

Code flow

A short-lived code crosses the browser; tokens use a back-channel exchange.

3

PKCE and state

PKCE binds the code to its client; state protects callback correlation.

MAKE THE DECISION

USE IT WHEN

Social login, enterprise SSO, integrations, delegated access, and mobile apps.

WATCH OUT FOR

Loose redirect URIs, token leakage, missing state/PKCE, and overbroad scopes.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

OAuth answers who may access an API; OIDC adds who authenticated to the client.

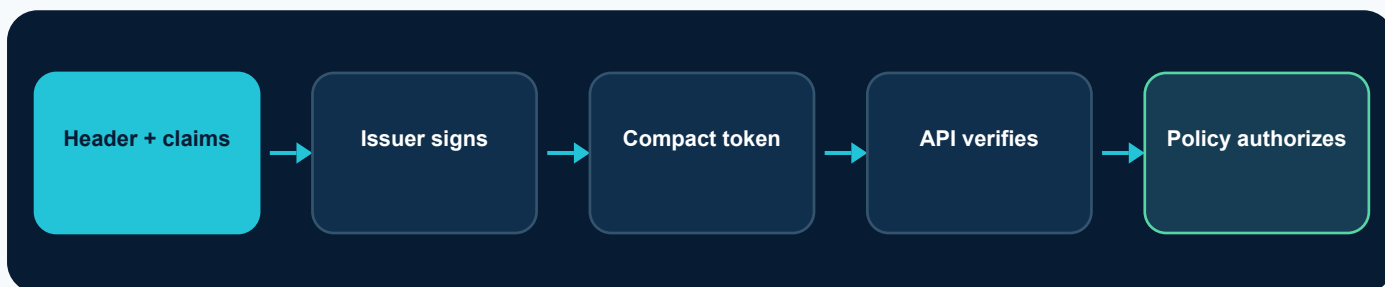
JWT

A compact signed envelope for claims exchanged between trust domains.

IN ONE MINUTE

A JWT contains Base64URL-encoded header and claims plus a signature. Receivers must verify the signature and relevant claims before trust.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Signed, not secret

Anyone holding a normal JWT can decode its payload; encryption is separate.

2

Claim validation

Check issuer, audience, expiry, not-before, subject, and app requirements.

3

Revocation gap

Stateless verification favors short lifetimes, rotation, and selective deny lists.

MAKE THE DECISION

USE IT WHEN

Short-lived access tokens, signed assertions, and portable claims between known parties.

WATCH OUT FOR

Algorithm confusion, sensitive payloads, weak keys, broad audiences, and long expiry.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

Decoding a JWT proves nothing; trust begins after signature and claim verification.

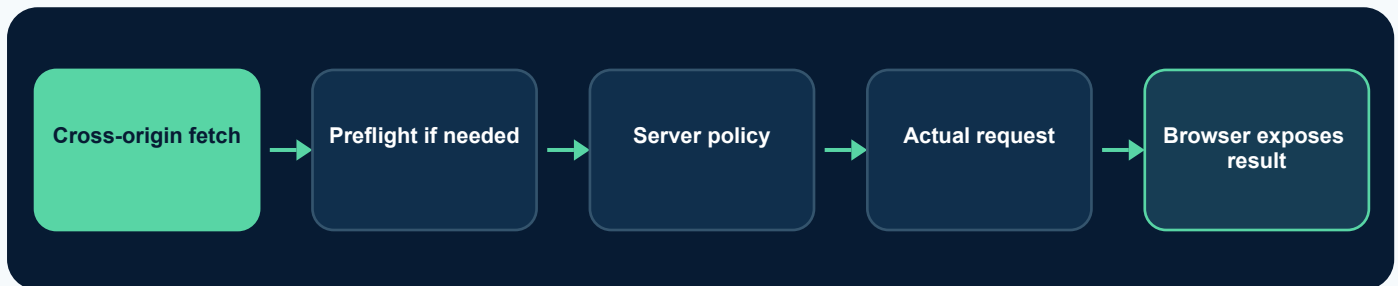
CORS

A browser rule controlling which origins may read cross-origin responses.

IN ONE MINUTE

CORS extends the same-origin policy with response headers. For non-simple requests, the browser sends an OPTIONS preflight first.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Origin

Scheme, host, and port determine whether browser contexts share an origin.

2

Preflight

OPTIONS asks whether a method and request headers are permitted.

3

Browser enforcement

curl and backend services ignore CORS, so it is not API authentication.

MAKE THE DECISION

USE IT WHEN

Browser applications that intentionally call an API on another trusted origin.

WATCH OUT FOR

Wildcard credentials, reflected origins, missing Vary, and confusing CORS with auth.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

CORS controls browser response reads; your API still needs real access control.

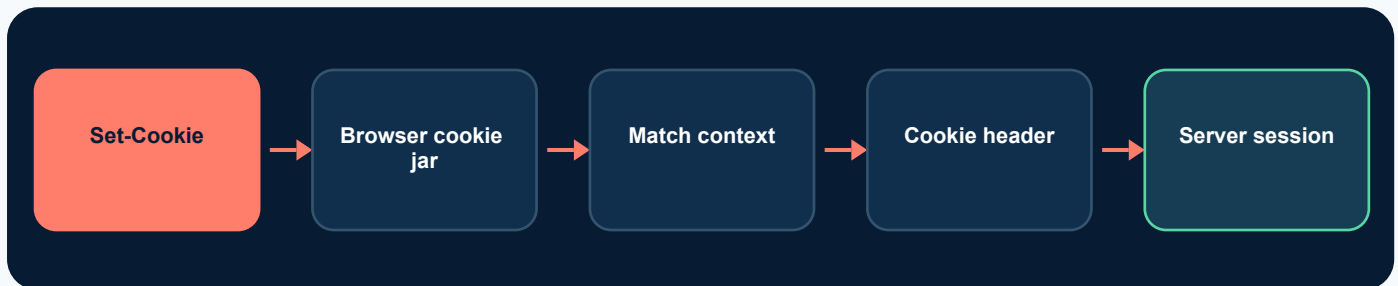
HTTP Cookies

Browser-managed state attached to requests by matching and security rules.

IN ONE MINUTE

A server stores a small pair with Set-Cookie. The browser later attaches it when domain, path, expiry, Secure, and SameSite rules allow.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

HttpOnly

Blocks JavaScript reads, reducing direct session theft through many XSS paths.

2

Secure

Allows the browser to send the cookie only over authenticated HTTPS.

3

SameSite and scope

SameSite, Domain, and Path limit when and where the cookie travels.

MAKE THE DECISION

USE IT WHEN

Session identifiers, preferences, CSRF token patterns, and small browser state.

WATCH OUT FOR

XSS, CSRF, broad Domain scope, missing flags, fixation, and oversized headers.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

Cookies are sent by browser policy, so attributes are part of authentication design.

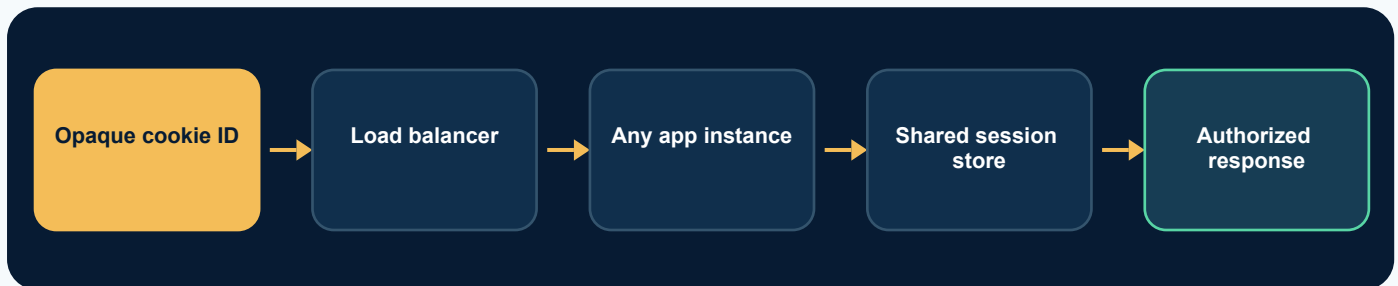
Session Management

Keep mutable login state behind an opaque browser identifier.

IN ONE MINUTE

A session cookie carries an unpredictable ID. Application instances look up identity, authorization, expiry, and risk state in a shared server-side store.

MINIMAL CONCEPT FLOW



CORE IDEAS

1

Strong identifiers

Use CSPRNG values with enough entropy; do not encode user meaning in the ID.

2

Expiration

Idle and absolute limits bound unattended and total session lifetime.

3

Rotation and revoke

Rotate after login or privilege change; index sessions for logout-all.

MAKE THE DECISION

USE IT WHEN

Browser login, immediate revocation, server-controlled state, and multi-device views.

WATCH OUT FOR

Fixation, stolen IDs, Redis failure, unlimited lifetime, and weak rotation.

PRODUCTION LENS

What fails? What retries? What state is shared? What can be cached?

REMEMBER

The cookie identifies the session; the server record decides if it is still trusted.